PHB 227
Statistical Learning and Multivariate Analysis

Classification

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Classification

What is a classification problem?

Classification

What is a classification problem?

Suppose Y has K categories, labeled $0, 1, \cdots, K-1$. Given X, we want a rule $\hat{f}(X)$ to classify an observation to one of the categories.

Can we use linear regression?

Methods

1. Logistic Regression
2. Linear Discriminant Analysis (LDA)
3. Quadratic Discriminant Analysis (QDA)
4. k-Nearest Neighbor (kNN), Trees, Random Forest, Support Vector Machines

Logistic Regression

- Suppose Y is a binary variable with values 0 and 1
- Logistic regression posits a (parametric) model for the probability distribution of Y.
- $p(X) = P(Y = 1|X_1, X_2, \cdots X_p) =$???

Logistic Regression

- Suppose Y is a binary variable with values 0 and 1
- Logistic regression posits a (parametric) model for the probability distribution of Y.
- $p(X) = P(Y = 1|X_1, X_2, \cdots X_p) =$???
- $p(X) = P(Y = 1|X_1, X_2, \cdots X_p) = \frac{exp(\beta_0+\beta_1 X_1+\beta_2 X_2 \cdots \beta_p X_p)}{1+exp(\beta_0+\beta_1 X_1+\beta_2 X_2 \cdots \beta_p X_p)}$
- Need to estimate βs. This is achieved, for e.g., by maximum likelihood. (Review MLE for logistic model.)

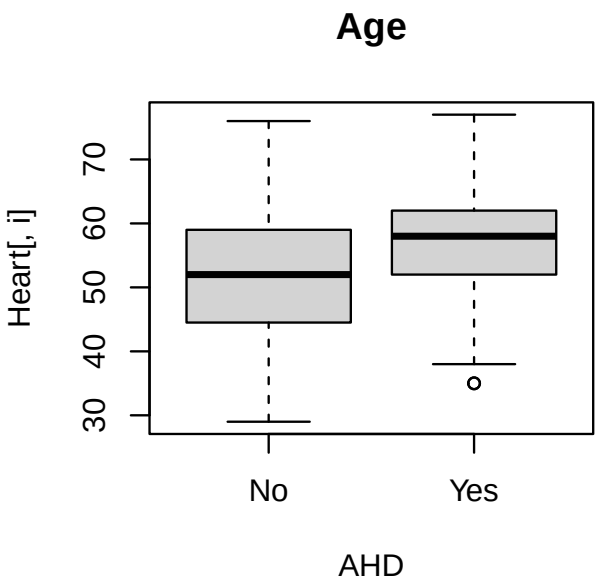
Heart data (ISLR)

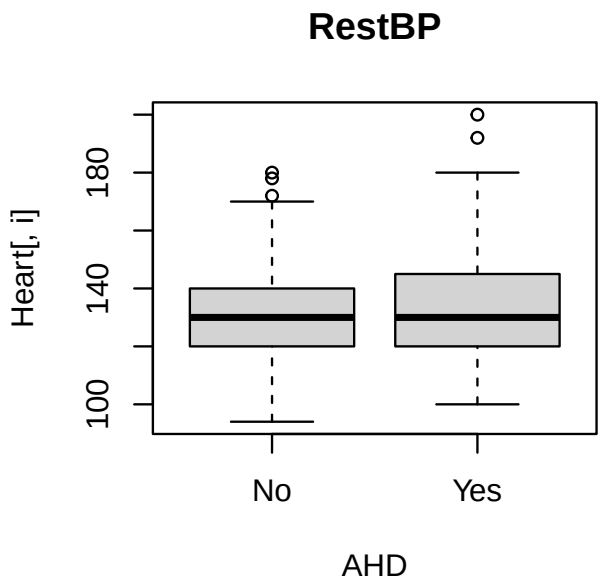
```
Heart <- read.csv("Heart.csv", header = TRUE, sep=",")
names(Heart)[-1]

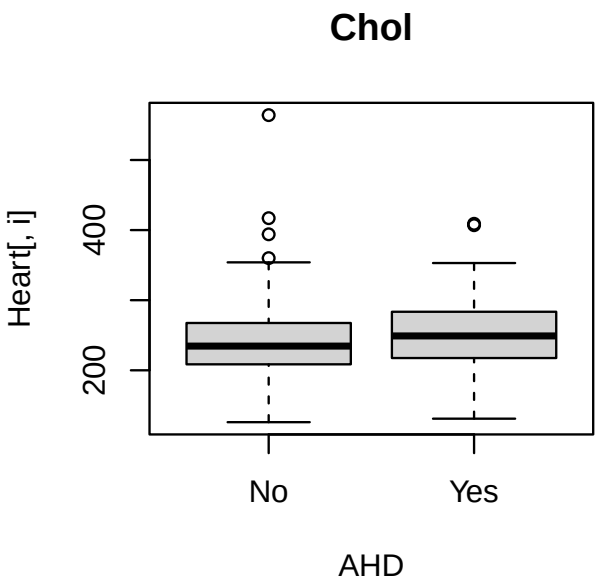
##  [1] "Age"       "Sex"       "ChestPain" "RestBP"
##  [5] "Chol"      "Fbs"       "RestECG"   "MaxHR"
##  [9] "ExAng"     "Oldpeak"   "Slope"     "Ca"
## [13] "Thal"      "AHD"

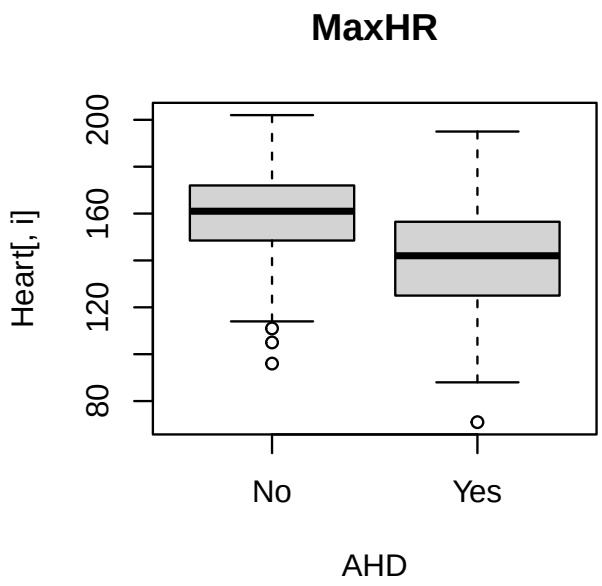
Heart$AHD <- as.factor(Heart$AHD)
summary(Heart[,c(2,3,5,6,9,15)])

##       Age             Sex             RestBP
##  Min.   :29.00   Min.   :0.0000   Min.   : 94.0
##  1st Qu.:48.00   1st Qu.:0.0000   1st Qu.:120.0
##  Median :56.00   Median :1.0000   Median :130.0
##  Mean   :54.44   Mean   :0.6799   Mean   :131.7
##  3rd Qu.:61.00   3rd Qu.:1.0000   3rd Qu.:140.0
##  Max.   :77.00   Max.   :1.0000   Max.   :200.0
##       Chol            MaxHR         AHD
##  Min.   :126.0   Min.   : 71.0   No :164
##  1st Qu.:211.0   1st Qu.:133.5   Yes:139
##  Median :241.0   Median :153.0
##  Mean   :246.7   Mean   :149.6
##  3rd Qu.:275.0   3rd Qu.:166.0
##  Max.   :564.0   Max.   :202.0
```


Age
Heart[, i]
30
40
50
60
70
No
Yes
AHD


RestBP
Heart[, i]
100
140
180
No
Yes
AHD


Chol
Heart[, i]
200
400
No
Yes
AHD


MaxHR
Heart[, i]
80
120
160
200
No
Yes
AHD

Logistic Regression application to Heart data (ISLR)

```
xnams <- c("Age", "Sex", "RestBP", "Chol", "MaxHR")
fmla <- as.formula(paste("as.factor(AHD) ~ ", paste(xnams, collapse= "+")))
logr.fit <- glm(fmla,data = Heart, family=binomial)
```

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	0.19	1.82	0.11	0.92
Age	0.01	0.02	0.79	0.43
Sex	1.82	0.34	5.35	0.00
RestBP	0.02	0.01	2.53	0.01
Chol	0.01	0.00	2.58	0.01
MaxHR	-0.05	0.01	-6.05	0.00

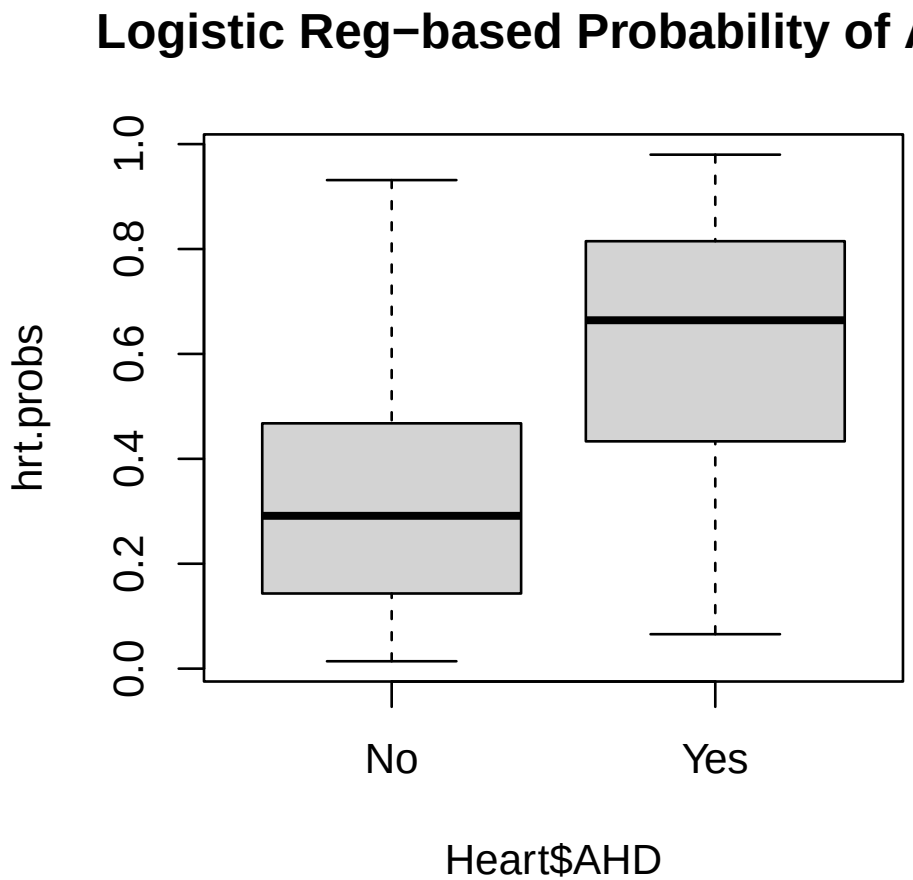
Table: Logistic Regression Output

Based on this model, what is the probability of AHD for a male subject 54 years of age, Rest BP of 100 mm Hg, Cholesterol 220 mg/dl, and MaxHR of 150?

Let us extract the probability distribution of AHD.

```
hrt.probs <- predict(logr.fit, type = "response")
summary(hrt.probs)

##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01412 0.23732 0.44031 0.45875 0.67499 0.97987
```



How does this help us classify?

We will classify as having AHD if the model-derived probability is > 0.5.
What is the justification for this?

We will classify as having AHD if the model-derived probability is > 0.5. What is the justification for this?

```
tbl <- table(Heart$AHD, hrt.probs>0.5)

crrcls <- (tbl[1,1] + tbl[2,2])/sum(tbl)

tbl

##
##       FALSE TRUE
##   No    125   39
##   Yes    51   88

crrcls

## [1] 0.7029703
```

Model Performance Metrics

	Predicted Class $\hat{Y}=0$	$\hat{Y}=1$	Total T
True $Y=0$	TN	FP	T^-
Class $Y=1$	FN	TP	T^+
Total	$\hat{T}^-$	$\hat{T}^+$	

Metric	Definition	Estimate	Synonyms
Sensitivity	$P(\hat{Y}=1\|Y=1)$	TP/T^+	True Pos Rate, Power
Specificity	$P(\hat{Y}=0\|Y=0)$	TN/T^-	1- Type 1 error True Neg rate
Pos. Pred. Value	$P(Y=1\|\hat{Y}=1)$	$TP/\hat{T}^+$	1 - FDR
Neg. Pred. Value	$P(Y=0\|\hat{Y}=0)$	$TN/\hat{T}^-$	

```
sens <- tbl[2,2]/(tbl[2,1] + tbl[2,2])
spec <- tbl[1,1]/(tbl[1,1] + tbl[1,2])
ppv <- tbl[2,2]/(tbl[1,2] + tbl[2,2])
npv <- tbl[1,1]/(tbl[1,1] + tbl[2,1])
```

```
##                             Accuracy.logr
## Correct Classification              0.703
## Sensitivity                         0.633
## Specificity                         0.762
## Positive Predictive Value           0.693
## Negative Predictive value           0.710
```

In the Heart example, how is the performance of the classifier affected if we classify as having AHD if the model-derived probability is > 0.3.? How is performance affected if we classify as having AHD if the model-derived probability is > 0.8?

Logistic Regression Classifier Rule

Given Y a binary variable with values 0 and 1, and a logistic regression model $P(Y=1|X_1, X_2, \cdots X_p) = \frac{exp(\beta_0+\beta_1 X_1+\beta_2 X_2 \cdots \beta_p X_p)}{1+exp(\beta_0+\beta_1 X_1+\beta_2 X_2 \cdots \beta_p X_p)}$, the classification rule is:

$$\text{Classify } \vec{X} \text{ as having } Y=1 \text{ if}$$
$$P(Y=1|\vec{X}) > 0.5$$
$$\text{equivalently}$$
$$odds(Y|\vec{X}) = \frac{P(Y=1|\vec{X})}{1-P(Y=1|\vec{X})} = exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2 \cdots \beta_p X_p) > 1$$
$$\text{equivalently}$$
$$ln\left(\frac{P(Y=1|\vec{X})}{1-P(Y=1|\vec{X})}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 \cdots \beta_p X_p > 0.$$

Discriminant Analysis

Assume two classes $Y = 1$ and $Y = 0$ with frequencies π_1 and π_0 respectively.

Suppose $\vec{X} = (X_1, X_2, \cdots X_p)$ has distribution dependent on Y, namely, $P(\vec{X}|Y = 0) = f_0(\vec{X})$ and $P(\vec{X}|Y = 1) = f_1(\vec{X})$.

By Bayes theorem, $P(Y = 0|\vec{X}) =$???

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$P(Y = 0|\vec{X}) = \frac{\pi_0 f_0(\vec{X})}{\pi_0 f_0(\vec{X}) + \pi_1 f_1(\vec{X})}$ and $P(Y = 1|\vec{X}) = \frac{\pi_1 f_1(\vec{X})}{\pi_0 f_0(\vec{X}) + \pi_1 f_1(\vec{X})}$

So we can extract probabilities. These are referred to as *posterior* probabilities.

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So we can extract probabilities. These are referred to as *posterior* probabilities.

Bayes optimal classifier rule states that we should classify to class $Y = k$ for which the posterior probability, $P(Y = k|\vec{X})$ is larger.

Linear Discriminant Analysis (LDA)

Suppose $\vec{X} = (X_1, X_2, \cdots X_p)$ are multivariate normal with distribution $\mathcal{N}(\mu_0, \Sigma)$ if $\vec{X}$ is from group 0, and $\mathcal{N}(\mu_1, \Sigma)$ if $\vec{X}$ is from group 1

$$P(Y = 0|\vec{X}) = \frac{\pi_0 f_0(\vec{X})}{\pi_0 f_0(\vec{X}) + \pi_1 f_1(\vec{X})} = \frac{\pi_0 \frac{1}{\sqrt{((2\pi)^p |\Sigma|)}} exp(-1/2(\vec{X}-\mu_0)^t \Sigma^{-1}(\vec{X}-\mu_0))}{\pi_0 \frac{1}{\sqrt{((2\pi)^p |\Sigma|)}} exp(-1/2(\vec{X}-\mu_0)^t \Sigma^{-1}(\vec{X}-\mu_0)) + \pi_1 \frac{1}{\sqrt{((2\pi)^p |\Sigma|)}} exp(-1/2(\vec{X}-\mu_1)^t \Sigma^{-1}(\vec{X}-\mu_1))}.$$

Similarly, we can write down $P(Y = 1|\vec{X})$.
By Bayes optimal classifier rule, we want to assign to whichever class has the higher posterior probability.
After some algebra ..... we get the following classification rule:

assign to class $k = 0$ or 1 for which the following is larger,

$$\delta_k(x) = x^t \Sigma^{-1} \mu_k - 0.5 \mu_k^t \Sigma^{-1} \mu_k + log(\pi_k).$$

The LDA classification rule is:

assign to class $k = 0$ or 1 for which the following is larger,

$$\delta_k(x) = x^t \Sigma^{-1} \mu_k - 0.5 \mu_k^t \Sigma^{-1} \mu_k + log(\pi_k)$$

Alternatively,

Assign to class 1 if

$$(x - 0.5(\mu_1 + \mu_0))^t \Sigma^{-1} (\mu_1 - \mu_0) + log \frac{\pi_1}{\pi_0} > 0$$

or equivalently

$$x^t \Sigma^{-1} (\mu_1 - \mu_0) - 0.5(\mu_1 + \mu_0)^t \Sigma^{-1} (\mu_1 - \mu_0) + log \frac{\pi_1}{\pi_0} > 0$$

Otherwise assign to class 0.

Note: This is equivalent to classifying to class $Y = 1$ if $P(Y = 1|\vec{X}) > 0.5$. Why?

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Otherwise assign to class 0.

Note: This is equivalent to classifying to class $Y = 1$ if $P(Y = 1|\vec{X}) > 0.5$. Why?

The *Bayes decision boundary* is the set of values of x for which $\delta_0(x) = \delta_1(x)$. What kind of surface is this in R^p-space?

In order to apply LDA, we need estimates of π_k, μ_k, Σ.

Use the proportions in each class in the (training) dataset to estimate π_k.

Given $\vec{X_1}, \vec{X_2}, \cdots, \vec{X_n}$ i.i.d random variables from $\mathcal{N}_p(\mu, \Sigma)$, we can obtain estimates of μ and Σ:

$$\hat{\mu} = \frac{1}{n}(\sum_{i=1}^{n} X_{i1}, \sum_{i=1}^{n} X_{i2}, \cdots, \sum_{i=1}^{n} X_{ip})$$

$$\hat{\Sigma} = \frac{1}{n}\sum_{i=1}^{n}(\vec{X_i} - \hat{\mu})(\vec{X_i} - \hat{\mu})^t = \frac{n-1}{n}S.$$

These are the MLEs of μ and Σ.

For LDA we assume the covariance matrix is the same in both classes. So use the pooled variance $S_{pooled} = \frac{n_1-1}{(n_1-1)+(n_2-1)}S_1 + \frac{n_2-1}{(n_1-1)+(n_2-1)}S_2$, which is an unbiased estimate of Σ.
What is the sample version of the LDA rule?

LDA application to Heart data

```
lda.fit <- lda(fmla,data = Heart)
round(lda.fit$prior,3)

##    No   Yes
## 0.541 0.459

round(lda.fit$means,3)

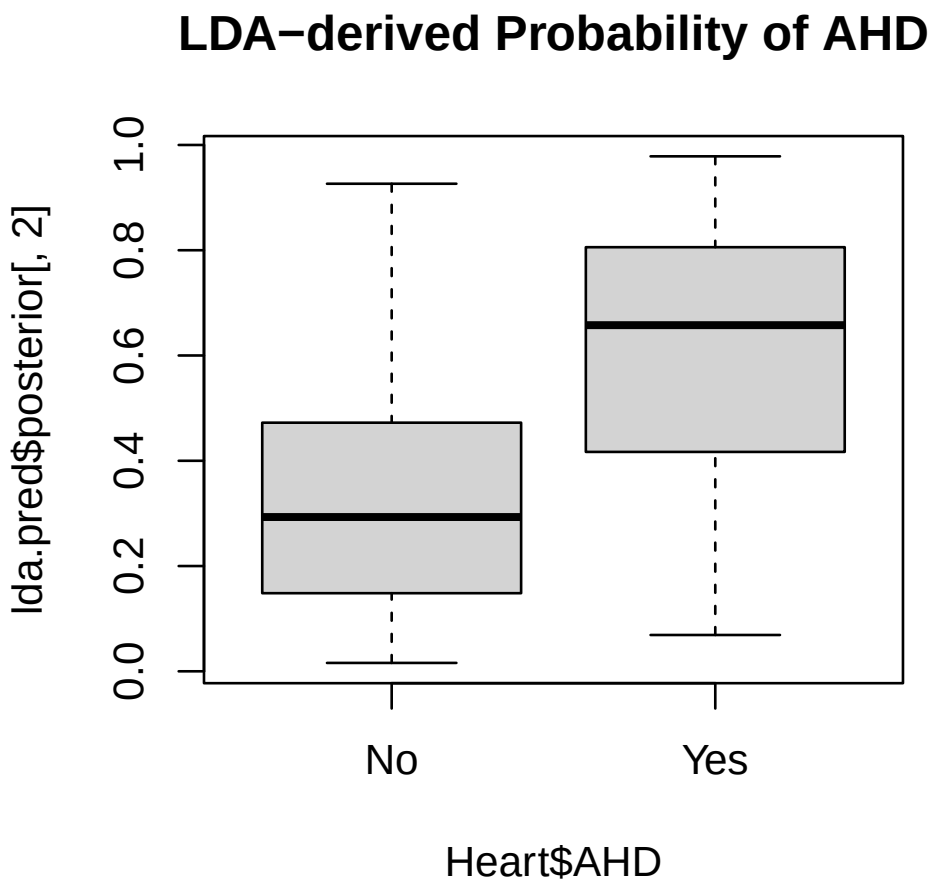
##        Age   Sex  RestBP    Chol   MaxHR
## No  52.585 0.561 129.250 242.640 158.378
## Yes 56.626 0.820 134.568 251.475 139.259

round(coef(lda.fit),3)

##           LD1
## Age     0.010
## Sex     1.401
## RestBP  0.016
## Chol    0.005
## MaxHR  -0.037
```

```
lda.pred <- predict(lda.fit,Heart)
#table(lda.pred£class, Heart£AHD)
summary(lda.pred$posterior[,2])  ##posterior prob of Y=1

##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01592 0.24236 0.42527 0.45493 0.66888 0.97841
```



```r
tbl <- table(Heart$AHD, lda.pred$posterior[,2]>0.5)

crrcls <- (tbl[1,1] + tbl[2,2])/sum(tbl)
sens <- tbl[2,2]/(tbl[2,1] + tbl[2,2])
spec <- tbl[1,1]/(tbl[1,1] + tbl[1,2])
ppv <- tbl[2,2]/(tbl[1,2] + tbl[2,2])
npv <- tbl[1,1]/(tbl[1,1] + tbl[2,1])

tbl
```

```
##
##       FALSE TRUE
##   No    130   34
##   Yes    52   87
```

```r
dd$Accuracy.lda <- c(crrcls,sens,spec,ppv,npv)
round(dd,3)
```

```
##                            Accuracy.logr Accuracy.lda
## Correct Classification             0.703        0.716
## Sensitivity                        0.633        0.626
## Specificity                        0.762        0.793
## Positive Predictive Value          0.693        0.719
## Negative Predictive value          0.710        0.714
```

LDA vs Logistic Regression

- Logistic regression fits a conditional model $P(Y|X)$; LDA fits a joint likelihood $P(Y, X) = \pi_0\phi(X, \mu_0, \Sigma) + \pi_1\phi(X, \mu_1, \Sigma)$. In other words, logistic regression ignores the distribution of X.
- LDA and logistic have linear classification rules. Classify $\vec{X}$ as having $Y = 1$ if
  - Logistic: $\beta_0 + \beta_1 X_1 + \beta_2 X_2 \cdots \beta_p X_p > 0$.
  - LDA: $x^t\Sigma^{-1}(\mu_1 - \mu_0) - 0.5(\mu_1 - \mu_0)^t\Sigma^{-1}(\mu_1 + \mu_0) + log\frac{\pi_1}{\pi_0} > 0$
- Methods differ in how the parameters of these rules are estimated
- LDA assumes multivariate Gaussian distribution for the predictors, so may do poorly if the predictors are non-Gaussian (e.g., categorical); so logistic regression is more robust.
- If the predictors are multivariate Gaussian then logistic regression is less efficient (i.e., parameter estimates have higher variance) because it does not use this information. This loss of efficiency can vary between 50% to 67% for "informative" classifiers. See Efron JASA 1975.

Another derivation of Bayes Optimal Classification Rule

See JW Section 11.2.
(Derived in class time-permitting)

LDA: Misclassification Error

Recall the LDA rule, namely,

$$\text{Assign to class 1 if}$$
$$R_1 : x^t\Sigma^{-1}(\mu_1 - \mu_0) - 0.5(\mu_1 - \mu_0)^t\Sigma^{-1}(\mu_1 + \mu_0) + log\frac{\pi_1}{\pi_0} > 0$$

$$\text{Assign to class 0 if}$$
$$R_0 : x^t\Sigma^{-1}(\mu_1 - \mu_0) - 0.5(\mu_1 - \mu_0)^t\Sigma^{-1}(\mu_1 + \mu_0) + log\frac{\pi_1}{\pi_0} \leq 0$$

Under the LDA rule, and assuming $\pi_1 = \pi_0$, misclassification error is $\Phi(-\Delta/2)$

where Φ is the cumulative distribution of a standard (univariate) Gaussian random variable, $\Delta^2 = (\mu_1 - \mu_0)^t\Sigma^{-1}(\mu_1 - \mu_0)$.

Let us derive this....

Note: Δ^2 is called the *Mahalanobis distance*.

Fisher's Linear Discriminant

Fisher arrived at LDA rule via a different argument.

- Take linear combinations of (multivariate) $\vec{X}$ to get projections $z = \vec{a}^T X$
- Find the linear combination, $\vec{a}$, that maximizes the separation of z's from group 0 and group 1, i.e., maximizes the ratio

$$\frac{(\vec{a}^T \mu_1 - \vec{a}^T \mu_0)^2}{\vec{a}^T \Sigma \vec{a}}$$

- The linear combination that achieves this maximal separation is $\vec{a} = c\Sigma^{-1}(\mu_1 - \mu_0)$, for nonzero scalar c.

Fisher's rule follows from the following fact.
Lemma: Given a positive definite $(p \times p)$ matrix W and a p-vector $\vec{d}$, the vector $\vec{a}$ which maximizes $\frac{(\vec{a}^T \vec{d})^2}{\vec{a}^T W \vec{a}}$ is given by $\vec{a} = cW^{-1}\vec{d}$, for nonzero scalar c. Furthermore the achieved maximum is $\vec{d}^T W^{-1} \vec{d}$.

Fisher's Linear Discriminant Rule

Set $\vec{a} = S_{pooled}^{-1}(\hat{\mu}_1 - \hat{\mu}_0)$, with $\hat{\mu}_1, \hat{\mu}_0, S_{pooled}$ equal to the sample estimates of μ_1, μ_0, Σ respectively.

Fisher's rule: Let $\hat{m} = 0.5(\hat{\mu}_1 - \hat{\mu}_0)^T S_{pooled}^{-1}(\hat{\mu}_1 + \hat{\mu}_0) = 0.5\vec{a}^T(\hat{\mu}_1 + \hat{\mu}_0)$,

Allocate $\vec{x_0}$ to class 1 if ${x_0}^T\vec{a} > \hat{m}$.
Allocate $\vec{x_0}$ to class 0 if ${x_0}^T\vec{a} \leq \hat{m}$.

Note: Fisher's rule does not assume multivariate normality of the populations. It does assume that the population covariance matrices are equal.

Fisher Discriminant

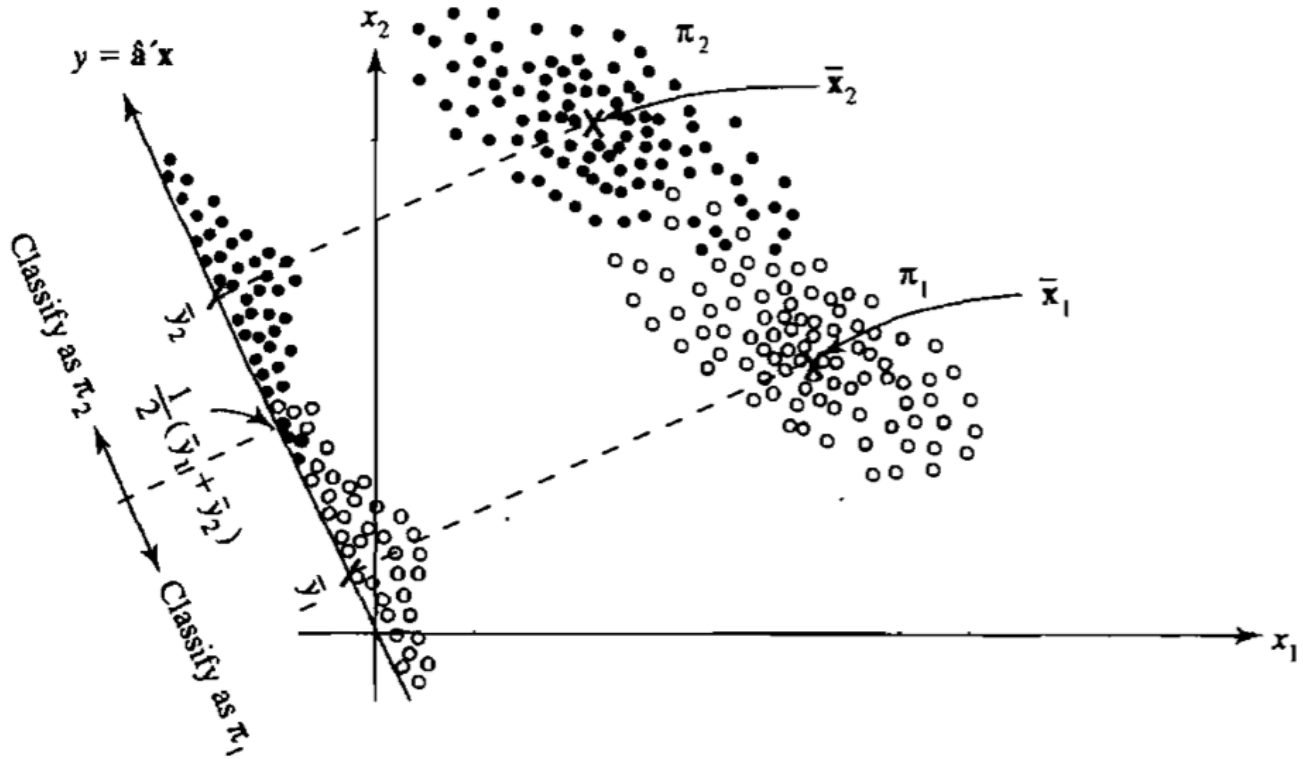


Figure 11.5 A pictorial representation of Fisher's procedure for two populations with $p = 2$.

reproduced from J & W

Quadratic Discriminant Analysis (QDA)

- Relax the LDA assumption of equal covariance matrices across classes.
- So now in the two class case, $\vec{X} = (X_1, X_2, \cdots X_p)$ are multivariate normal with distribution
  $\mathcal{N}(\mu_0, \Sigma_0)$ if $\vec{X}$ is from group 0, and $\mathcal{N}(\mu_1, \Sigma_1)$ if $\vec{X}$ is from group 1 where Σ_k is the covariance matrix for class k.
- The QDA classification rule assigns to class k for which the following is larger:
  $-0.5(x-\mu_k)^t\Sigma_k^{-1}(x-\mu_k) - 0.5\log|\Sigma_k| + \log(\pi_k) =$
  $-0.5x^t\Sigma_k^{-1}x - 0.5\log|\Sigma_k| + x^t\Sigma_k^{-1}\mu_k - 0.5\mu_k^t\Sigma_k^{-1}\mu_k + \log(\pi_k)$
- Follows from using Bayes theorem and Bayes Optimal Classification rule

QDA classification rule

Assign to class k for which the following is larger:

$$-0.5x^t\Sigma_k^{-1}x - 0.5\log|\Sigma_k| + x^t\Sigma_k^{-1}\mu_k - 0.5\mu_k^t\Sigma_k^{-1}\mu_k + \log(\pi_k)$$

or equivalently, assign to class 1 if

$$-0.5x^t(\Sigma_1^{-1} - \Sigma_0^{-1})x + x^t(\Sigma_1^{-1}\mu_1 - \Sigma_0^{-1}\mu_0) - C + \log(\frac{\pi_1}{\pi_0}) \geq 0,$$

where $C = 0.5\log\frac{|\Sigma_1|}{|\Sigma_0|} + 0.5(\mu_1^t\Sigma_1^{-1}\mu_1 - \mu_0^t\Sigma_0^{-1}\mu_0)$.

What kind of surface in R^p is the QDA boundary?

QDA application to Heart data

```
qda.fit <- qda(fmla,data = Heart)
round(qda.fit$prior,3)

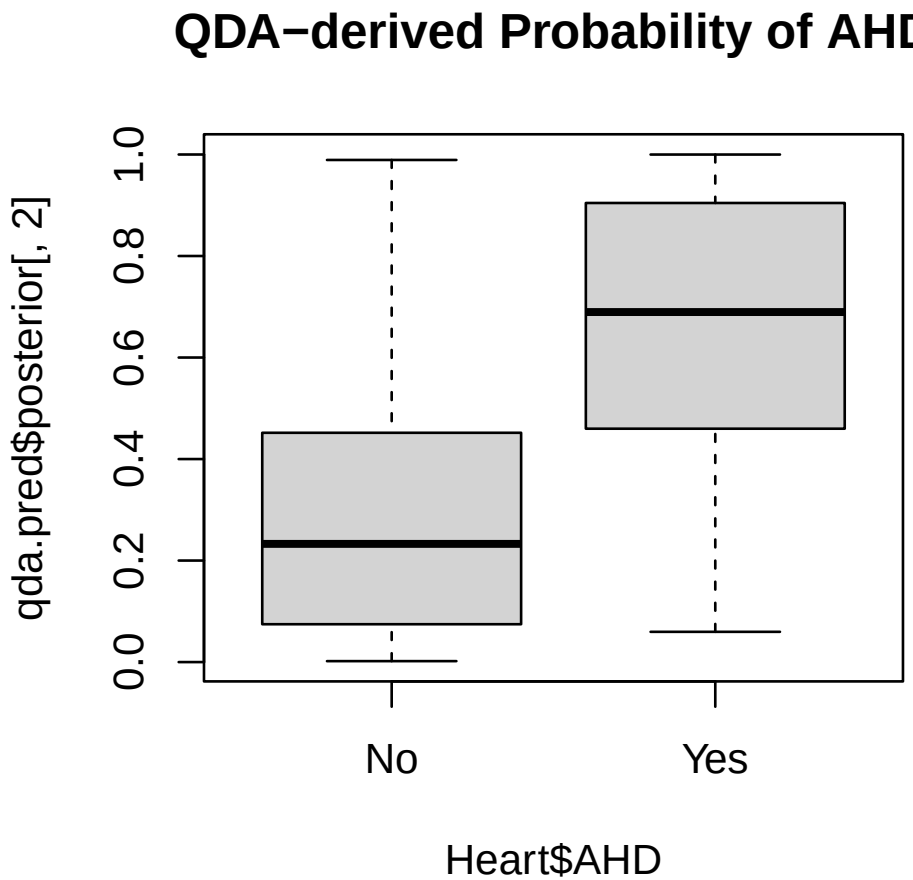
##    No   Yes
## 0.541 0.459

round(qda.fit$means,3)

##        Age   Sex  RestBP    Chol   MaxHR
## No  52.585 0.561 129.250 242.640 158.378
## Yes 56.626 0.820 134.568 251.475 139.259
```

```
qda.pred <- predict(qda.fit,Heart)
summary(qda.pred$posterior[,2])  ##posterior prob of Y=1

##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
## 0.001875 0.161925 0.433004 0.453526 0.707624 0.999902
```



```r
tbl <- table(Heart$AHD, qda.pred$posterior[,2]>0.5)
tbl
```

```
##
##       FALSE TRUE
##   No    132   32
##   Yes    41   98
```

```r
round(dd,3)
```

```
##                            Accuracy.logr Accuracy.lda
## Correct Classification             0.703        0.716
## Sensitivity                        0.633        0.626
## Specificity                        0.762        0.793
## Positive Predictive Value          0.693        0.719
## Negative Predictive value          0.710        0.714
##                            Accuracy.qda
## Correct Classification            0.759
## Sensitivity                       0.705
## Specificity                       0.805
## Positive Predictive Value         0.754
## Negative Predictive value         0.763
```

k-Nearest Neighbors

Suppose Y is a binary variable with values 0 and 1. Suppose our data set consists of n pairs $(x_1, y_1), (x_2, y_2), \cdots (x_n, y_n)$. For a datum x, let $\mathcal{N}_k(x)$ consist of a set of k x_is that are "closest" to x (in Euclidean distance). The predicted probability of Y is derived as

$$P(Y = 1|x) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(x)} I(y_i = 1).$$

As before, we will predict $\hat{Y} = 1$ if the derived probability is > 0.5.

K-NN is a highly nonparametric approach. The choice of k determines the level of flexibility: $k = 1$ is very flexible, as k increases the procedure is less flexible.

k-NN: Application to Heart data

$k = 5$:

```
knn.pred <- knn(Heart[,xnams], Heart[,xnams],Heart$AHD, k=5)
tbl <- table(Heart$AHD, knn.pred)
round((tbl[1,1] + tbl[2,2])/sum(tbl),2)

## [1] 0.77
```

```
##                            Accuracy.logr Accuracy.lda
## Correct Classification             0.703        0.716
## Sensitivity                        0.633        0.626
## Specificity                        0.762        0.793
## Positive Predictive Value          0.693        0.719
## Negative Predictive value          0.710        0.714
##                            Accuracy.qda Accuracy.knn5
## Correct Classification            0.759         0.766
## Sensitivity                       0.705         0.719
## Specificity                       0.805         0.805
## Positive Predictive Value         0.754         0.758
## Negative Predictive value         0.763         0.772
```

k-NN: Application to Heart data

$k = 1$:

```
knn.pred <- knn(Heart[,xnams], Heart[,xnams],Heart$AHD, k=1)
tbl <- table(Heart$AHD, knn.pred)
(tbl[1,1] + tbl[2,2])/sum(tbl)

## [1] 1
```

$k = 1$, predict on "new" set:

```
knn.pred <- knn(Heart[1:230,xnams], Heart[231:303,xnams],Heart[1:230,15], k=1)
tbl <- table(knn.pred,Heart$AHD[231:303])
round((tbl[1,1] + tbl[2,2])/sum(tbl),2)

## [1] 0.56
```

$k = 5$, predict on "new" set:

```
## [1] 0.66
```

Comparing methods

Suppose you did a simulation in which data were generated according to the following scenarios with $p = 2$ predictors and two classes (groups). What ordering of the performance of logistic, lda, qda and k-NN classifiers might you expect?

1. The predictors X_1 and X_2 are uncorrelated Gaussian random variables with the same variance but a different mean in each class.
2. Same as Scenario 1, except that within each class, the two predictors have a correlation of -0.5.
3. Same as scenario 1, but now X_1 and X_2 are generated from a t-distribution
4. The predictors X_1 and X_2 are random Gaussian variables with a correlation of -0.5 in the first class and 0.5 in the second class.
5. Within each class the predictors are uncorrelated Gaussian. But the response is generated from a logistic function using ${X_1}^2$, ${X_2}^2$, and $X_1 X_2$ as predictors.

Multiclass (> 2) classification

Assume K classes $Y = 0, 1, \cdots K-1$ with frequencies $\pi_0, \pi_1, \cdots, \pi_{K-1}$ respectively, and p predictors, $\vec{X} = (X_1, X_2, \cdots X_p)$. We can generalize the previous classifiers.

Multinomial Logistic Regression

$$P(Y = l|X_1, X_2, \cdots X_p) = \frac{exp(\beta_{0l} + \beta_{1l}X_1 + \beta_{2l}X_2 \cdots \beta_{pl}X_p)}{1 + \sum_{k=1}^{K-1} exp(\beta_{0k} + \beta_{1k}X_1 + \beta_{2k}X_2 \cdots \beta_{pk}X_p)}$$
$$\text{for } l = 1, 2, \cdots, K-1$$

$$P(Y = 0|X_1, X_2, \cdots X_p) = \frac{1}{1 + \sum_{k=1}^{K-1} exp(\beta_{0k} + \beta_{1k}X_1 + \beta_{2k}X_2 \cdots \beta_{pk}X_p)}$$

Multiclass Discriminant Analysis

Suppose $\vec{X}$ has distribution dependent on Y, namely, $P(\vec{X}|Y = k) = f_k(\vec{X})$. By Bayes theorem, $P(Y = l|\vec{X}) = \frac{\pi_l f_l(\vec{X})}{\sum_{k=0}^{K-1} \pi_k f_k(\vec{X})}$.

As before the classification rule assigns to the class k for which $P(Y = k|\vec{X})$ is largest.

Multiclass (> 2) classification: Logistic Regression

Assume K classes $Y = 0, 1, \cdots K-1$ with frequencies $\pi_0, \pi_1, \cdots, \pi_{K-1}$ respectively, and p predictors, $\vec{X} = (X_1, X_2, \cdots X_p)$.

$$p_l = P(Y = l|\vec{X}) = \frac{exp(\beta_{0l} + \beta_{1l}X_1 + \beta_{2l}X_2 \cdots \beta_{pl}X_p)}{1 + \sum_{k=1}^{K-1} exp(\beta_{0k} + \beta_{1k}X_1 + \beta_{2k}X_2 \cdots \beta_{pk}X_p)}$$
$$\text{for } l = 1, 2, \cdots, K-1$$

$$p_0 = P(Y = 0|\vec{X}) = \frac{1}{1 + \sum_{k=1}^{K-1} exp(\beta_{0k} + \beta_{1k}X_1 + \beta_{2k}X_2 \cdots \beta_{pk}X_p)}$$

Multiclass Logistic Classifier Rule

Assign $\vec{X}$ to class l for which p_l $(l = 0, 1, \cdots, K-1)$ is maximum
i.,e, assign $\vec{X}$ to class l if $p_l \geq p_r$ for all r
i.e., identify class $l > 0$ for which
$(\beta_{0l} + \beta_{1l}X_1 + \beta_{2l}X_2 \cdots \beta_{pl}X_p \geq \beta_{0r} + \beta_{1r}X_1 + \beta_{2r}X_2 \cdots \beta_{pr}X_p)$ for all r,
assign X to class l if, in addition, $\beta_{0l} + \beta_{1l}X_1 + \beta_{2l}X_2 \cdots \beta_{pl}X_p \geq 0$,
else assign to class 0.

Multiclass (> 2) classification: LDA and QDA

Suppose $\vec{X}$ has distribution dependent on Y, namely, $P(\vec{X}|Y = k) = f_k(\vec{X})$. By Bayes theorem,

$$p_l = P(Y = l|\vec{X}) = \frac{\pi_l f_l(\vec{X})}{\sum_{k=0}^{K-1} \pi_k f_k(\vec{X})}.$$

As before the classification rule assigns to the class k for which $p_k = P(Y = k|\vec{X})$ is largest. Now assume f_k is multivariate normal, then we can derive LDA and QDA rules.

Multiclass (> 2) classification: LDA and QDA

Suppose $\vec{X}$ has distribution dependent on Y, namely, $P(\vec{X}|Y = k) = f_k(\vec{X})$. By Bayes theorem,

$$p_l = P(Y = l|\vec{X}) = \frac{\pi_l f_l(\vec{X})}{\sum_{k=0}^{K-1} \pi_k f_k(\vec{X})}.$$

As before the classification rule assigns to the class k for which $p_k = P(Y = k|\vec{X})$ is largest. Now assume f_k is multivariate normal, then we can derive LDA and QDA rules.
Let $\delta_k(\vec{x}) = -0.5(\vec{x} - \mu_k)^t \Sigma_k^{-1}(\vec{x} - \mu_k) - 0.5 log|\Sigma_k| + log(\pi_k)$.
Assign $\vec{x}$ to the class k for which $\delta_k(\vec{x})$ is largest
i.e., Assign $\vec{x}$ to class l if $\delta_l(\vec{x}) \geq \delta_k(\vec{x})$ for all k or equivalently

Multiclass (> 2) classification: LDA and QDA

Suppose $\vec{X}$ has distribution dependent on Y, namely, $P(\vec{X}|Y = k) = f_k(\vec{X})$. By Bayes theorem,

$$p_l = P(Y = l|\vec{X}) = \frac{\pi_l f_l(\vec{X})}{\sum_{k=0}^{K-1} \pi_k f_k(\vec{X})}.$$

As before the classification rule assigns to the class k for which $p_k = P(Y = k|\vec{X})$ is largest. Now assume f_k is multivariate normal, then we can derive LDA and QDA rules.
Let $\delta_k(\vec{x}) = -0.5(\vec{x} - \mu_k)^t \Sigma_k^{-1}(\vec{x} - \mu_k) - 0.5 log|\Sigma_k| + log(\pi_k)$.
Assign $\vec{x}$ to the class k for which $\delta_k(\vec{x})$ is largest
i.e., Assign $\vec{x}$ to class l if $\delta_l(\vec{x}) \geq \delta_k(\vec{x})$ for all k or equivalently
Multiclass QDA classifier rule: Assign $\vec{x}$ to class l if
$-0.5\vec{x}^t(\Sigma_l^{-1} - \Sigma_k^{-1})\vec{x} + \vec{x}^t(\Sigma_l^{-1}\mu_l - \Sigma_k^{-1}\mu_k) - 0.5(\mu_l^t \Sigma_l^{-1}\mu_l + \mu_k^t \Sigma_k^{-1}\mu_k) - 0.5 log \frac{|\Sigma_l|}{|\Sigma_k|} + log \frac{\pi_l}{\pi_k} \geq 0$ for all k

Multiclass (> 2) classification: LDA and QDA

Suppose $\vec{X}$ has distribution dependent on Y, namely, $P(\vec{X}|Y = k) = f_k(\vec{X})$. By Bayes theorem,

$$p_l = P(Y = l|\vec{X}) = \frac{\pi_l f_l(\vec{X})}{\sum_{k=0}^{K-1} \pi_k f_k(\vec{X})}.$$

As before the classification rule assigns to the class k for which $p_k = P(Y = k|\vec{X})$ is largest. Now assume f_k is multivariate normal, then we can derive LDA and QDA rules.
Let $\delta_k(\vec{x}) = -0.5(\vec{x} - \mu_k)^t \Sigma_k^{-1}(\vec{x} - \mu_k) - 0.5 log|\Sigma_k| + log(\pi_k)$.
Assign $\vec{x}$ to the class k for which $\delta_k(\vec{x})$ is largest
i.e., Assign $\vec{x}$ to class l if $\delta_l(\vec{x}) \geq \delta_k(\vec{x})$ for all k or equivalently
Multiclass QDA classifier rule: Assign $\vec{x}$ to class l if
$-0.5\vec{x}^t(\Sigma_l^{-1} - \Sigma_k^{-1})\vec{x} + \vec{x}^t(\Sigma_l^{-1}\mu_l - \Sigma_k^{-1}\mu_k) - 0.5(\mu_l^t\Sigma_l^{-1}\mu_l + \mu_k^t\Sigma_k^{-1}\mu_k) - 0.5 log\frac{|\Sigma_l|}{|\Sigma_k|} + log\frac{\pi_l}{\pi_k} \geq 0$ for all k
Multiclass LDA classifier rule: Assume $\Sigma_k = \Sigma\ \forall k$. Assign $\vec{x}$ to class l if
$\vec{x}^t\Sigma^{-1}(\mu_l - \mu_k) - 0.5(\mu_l^t\Sigma^{-1}\mu_l + \mu_k^t\Sigma^{-1}\mu_k) + log\frac{\pi_l}{\pi_k} \geq 0$ for all k.

Multiclass LDA

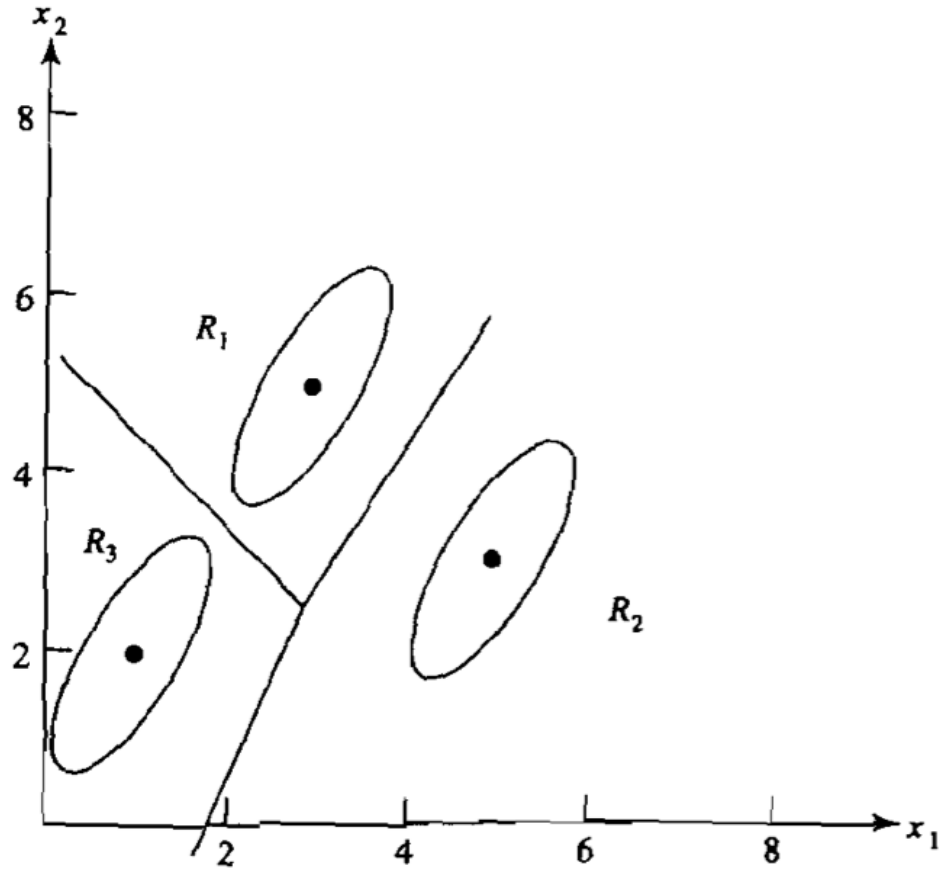


Figure 11.10 The classification regions R_1, R_2, and R_3 for the linear minimum TPM rule ($p_1 = \frac{1}{4}, p_2 = \frac{1}{2}, p_3 = \frac{1}{4}$).

reproduced from J & W

QDA vs LDA

How many parameters need to be estimated?

With p predictors, and since we only need to consider $\delta_k(x) - \delta_0(x)$,

- LDA has linear boundary: a plane with p predictors has p coefficients $+1$ intercept, giving a total of $(K-1)(p+1)$ parameters;
- QDA has a quadratic boundary: a quadratic with p predictors has $p(p+1)/2$ order 2 terms, p linear terms, 1 intercept, so a total of $(K-1)(\frac{p^2+3p}{2}+1)$ total parameters.

So, LDA is less flexible, hence more biased but has less variance. Conversely, QDA is more flexible, so less biased but has higher variance (many more parameters to estimate than LDA).

If n is large and p is small, QDA is better than LDA; else LDA is better.

Multivariate Normal Distribution

Given a random vector, $\vec{X} = (X_1, X_2, \cdots X_p)$, the p-dimensional multivariate normal density is given by

$$\frac{1}{\sqrt{((2\pi)^p|\Sigma|)}} exp(-1/2(\vec{X} - \mu)^t \Sigma^{-1}(\vec{X} - \mu))$$

μ is a $p \times 1$ vector, Σ is the variance-covariance matrix of $\vec{X}$. Σ is symmetric, and we will assume it is positive definite.

We denote this distribution by $\mathcal{N}_p(\mu, \Sigma)$

Some useful notes about Multivariate Normal Distributions

Let $\vec{X} \sim \mathcal{N}_p(\mu, \Sigma)$, $|\Sigma| > 0$,

1. The contours of constant density of $\mathcal{N}_p(\mu, \Sigma)$, are ellipsoids defined by $\vec{X}$ such that
$$(\vec{X} - \mu)^t \Sigma^{-1} (\vec{X} - \mu) = c^2.$$
These ellipsoids are centered at μ and have axes in the direction $\vec{e_i}$, with lengths proportional to $\sqrt{\lambda_i}$ where $\Sigma \vec{e_i} = \lambda_i \vec{e_i}$ for $i = 1, 2 \cdots p$.
2. If $\vec{X}$ is distributed $\mathcal{N}_p(\mu, \Sigma)$, then any linear combination $a^t \vec{X} = a_1 X_1 + a_2 X_2 + \cdots a_p X_p$ is distributed $\mathcal{N}(a^t \mu, a^t \Sigma a)$.
3. $(\vec{X} - \mu)^t \Sigma^{-1} (\vec{X} - \mu)$ is distributed as χ^2_p, the chi-square distribution with p degrees of freedom
4. The $\mathcal{N}_p(\mu, \Sigma)$, distribution assigns probability $1 - \alpha$ to the ellipsoid $\{\vec{X} : (\vec{X} - \mu)^t \Sigma^{-1} (\vec{X} - \mu)) \leq \chi^2_p(\alpha)\}$, where $\chi^2_p(\alpha)$ denotes the upper (100α) percentile of the χ^2_p distribution.

Given $\vec{X_1}, \vec{X_2}, \cdots, \vec{X_n}$ i.i.d random variables from $\mathcal{N}_p(\mu, \Sigma)$, we can obtain estimates of μ and Σ:

$$\hat{\mu} = \frac{1}{n}(\sum_{i=1}^{n} X_{i1}, \sum_{i=1}^{n} X_{i2}, \cdots, \sum_{i=1}^{n} X_{ip})$$

$$\hat{\Sigma} = \frac{1}{n}\sum_{i=1}^{n}(\vec{X_i} - \hat{\mu})(\vec{X_i} - \hat{\mu})^t = \frac{n-1}{n}S.$$

In fact, $\hat{\mu}$ and $\hat{\Sigma}$ (or S) are the *sufficient statistics* for the multivariate normal distribution $\mathcal{N}_p(\mu, \Sigma)$.

Given $\vec{X_1}, \vec{X_2}, \cdots, \vec{X_n}$ i.i.d random variables from $\mathcal{N}_p(\mu, \Sigma)$, we can obtain estimates of μ and Σ:

$$\hat{\mu} = \frac{1}{n}(\sum_{i=1}^{n} X_{i1}, \sum_{i=1}^{n} X_{i2}, \cdots, \sum_{i=1}^{n} X_{ip})$$

$$\hat{\Sigma} = \frac{1}{n}\sum_{i=1}^{n}(\vec{X_i} - \hat{\mu})(\vec{X_i} - \hat{\mu})^t = \frac{n-1}{n}S.$$

In fact, $\hat{\mu}$ and $\hat{\Sigma}$ (or S) are the *sufficient statistics* for the multivariate normal distribution $\mathcal{N}_p(\mu, \Sigma)$.

Furthermore,

1. $\hat{\mu}$ is distributed $\mathcal{N}_p(\mu, \frac{1}{n}\Sigma)$,
2. $(n-1)S$ is distributed as a Wishart random matrix with $n-1$ d.f.
3. $\hat{\mu}$ and S are independent.

Review Questions

1. Is logistic regression parametric or non-parametric?
2. In logistic regression, what function of Y are we modeling?
3. After we fit a logistic regression model, how would we classify a "new" data point if our goal is to minimize misclassification?
4. What is LDA?
5. What are similarities between logistic regression and LDA?
6. What are some differences between logistic regression and LDA?
7. What is the theoretical error rate of LDA?
8. What is Fisher's discriminant?
9. What is QDA?
10. How are LDA and QDA different?